MATHILDE **SIMONI**

Master's Student in Computational Sciences

CONTACT

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SKILLS

Languages English (fluent), French (native), Italian (intermediate), Spanish (elementary)

Coding languages Python, C++, MATLAB, Julia, R

Tools Git, ML tools (Scikit Learn, Pandas library), Blender, Ansys Fluent, MonolixSuite, FeniCSx library

Tech competencies data science, HPC, NLME modeling, numerical methods (finite difference, FEM), software development

AWARDS

Bending Spoons "Women in Computer Science Scholarship" recipient July 2024
Award winner at the *Advanced Computer Graphics Competition* at EPFL June 2024

LEADERSHIP AND COMMUNITY INVOLVMENT

Tutoring in Mathematics, Physics, and Computer Science 2022 – 2025

Member of the EPS and SE groups in the EPFL Spacecraft team 2023 – 2024

Mentor for the weSTEM Mentorship Program (volunteering) Fall 2022

Track & Field Intercollegiate team member 2019 - 2022

EDUCATION

EPFL (École Polytechnique Fédérale de Lausanne)

Lausanne, Switzerland | 2023-2026

Master of Science (MSc) in Computational Science and Engineering

Courses: Numerical Approximations to PDEs (FEM), Computer Simulations of Physical Systems, Advanced Numerical Analysis, Numerical Flow Simulation

New York University Abu Dhabi

Abu Dhabi, UAE | 2019-2023

Bachelor of Science in Mathematics and Computer Science (GPA of 3.937)

NYU Founders' Day Award and Cum Laude Latin Honors (University Honors Scholar)

Courses: Math Modeling, Computational Physics, Machine Learning, Big Data

Lycée International de Valbonne

Sophia-Antipolis, France | 2016-2019

Baccalauréat (French National Diploma) & Esame di Stato (Italian Diploma)

Scientific stream with specialization in Mathematics, Highest Honors (19.73/20)

ACADEMIC EXPERIENCE

A FULLY IMPLICIT SOLVER for LEMs (master's thesis)

EPFL CHANGE lab | Fall 2025

Developed a fully-implicit numerical solver for a transport-limited Landscape Evolution Model, a nonlinear system of PDEs coupling surface elevation and water dynamics. Explored alternative water velocity formulations to address ill-posedness and implemented adaptive time-stepping, regularization, and Schur-complement preconditioning to improve Newton solver convergence. Assessed solver performance on steady-state 1D and 2D landscapes.

CUBESAT DIGITAL TWIN (semester project)

EPFL Spacecraft team | Fall 2024

Implemented from scratch a Python-based simulation framework providing tools for the modeling and simulations of the CHESS CubeSat. It includes orbit propagation, as well as the EPS, Telecom, and ADCS subsystems.

PARALLELIZATION OF NYSTRÖM ALGORITHM

EPFL (HPC class) | Fall 2024

Implemented in parallel the randomized Nyström algorithm to compute rank-k approximations of positive semi-definite matrices. Investigated the numerical stability and parallel performance using 3 datasets and 2 sketching matrices (Gaussian and SHRT).

FEM SOLUTION OF A SEMILINEAR ELLIPTIC EQUATION

EPFL (Numerical Approximations to PDEs class) | Spring 2024

Developed a Finite Element Method code in Python and used 3 iterative solvers; the fixed-point iteration, Anderson's acceleration, and Newton's method, to solve a nonlinear problem.

STOMMEL'S MODEL OF THE THERMOHALINE CIRCULATION

NYU Abu Dhabi (Mathematical Modeling class) | Spring 2023

Analyzed Stommel's model of ocean circulation, identified equilibrium states, and studied their stability using linearization methods. Simulated the impact of Greenland ice sheet melt in the context of climate change using bifurcation theory.

WORK EXPERIENCE

MODELER INTERN

LYO-X, Basel, Switzerland | Spring 2025 (6 months)

Conducted research on antibody uptake modeling in solid tumors for cancer immunotherapy. Specifically implemented and extended 2 mechanistic pharmacokinetic models using MonolixSuite and R and validated them with collected data and literature-based parameter estimates.

STUDENT ASSISTANT IN THE MARS RESEARCH GROUP

Center for Space Science of NYU Abu Dhabi | 2022-2023

Analyzed temperature data from different NASA rovers on the planet's surface and conducted research to visualize and characterize auroras on Mars using data captured by the EMM mission. Published two papers as a co-author, available at doi.org/10.1093/mnrasl/slac094 and doi.org/10.1140/epjd/s10053-022-00566-5.